

Sound as the Mark of a Pervasive Medium

A Quantitative Test of a Vaiśeṣika Field Ontology

Against Greek, Chinese, and Buddhist Controls

and Maxwell Electromagnetism

Physics and Astronomy

Arjun Shah

Stratford Preparatory Milpitas
Independent Research / RISHI-Q
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Abstract

Popular claims that classical Indian thought anticipated electromagnetism or quantum physics usually rest on vague metaphors (ether, vibration, unity). This study tests a different, under-researched doctrine: in Kaṇāda's Vaiśeṣika system, ākāśa is a pervasive substance whose distinctive mark is sound (śabda), while light and heat belong to a separate substance (tejas).

Using GRETEL primary Sanskrit (no OCR), a nine-item theory checklist recovered 9/9 propositions. Commentarial replication in Praśastapāda recovered 6/6. A five-item Maxwell confrontation scored 0/5. On a frozen six-feature rubric comparing Vaiśeṣika, Lucretius, Plato's *Timaeus*, the Dao De Jing, the Dhammapada, and Maxwell electromagnetism, only Vaiśeṣika scored $R2 = 1$ (sound specially tied to the pervasive medium). Under a descriptive fair-coin null for $R2$ across six traditions, $P = 1/64 = 0.0156$.

The doctrine is historically sharp and computationally attested, but structurally incompatible with Maxwell unification. The contribution is an open multi-civilization falsification package—not a claim of ancient electromagnetic discovery.

1. Introduction

1.1 Motivation and Problem Statement

Claims that Sanskrit philosophical texts “anticipated” modern energy fields, electromagnetism, or quantum mechanics circulate widely in popular and semi-scholarly literature [Capra, 1975, Restivo, 1983]. Typical upgrades include: ether → quantum vacuum; vibration → quantum oscillation; unity → entanglement; ākāśa → electromagnetic field. These upgrades are culturally interesting as reception history, but they are scientifically soft: once modern physics is known, almost any metaphysical vocabulary can be reinterpreted until it sounds contemporary. A comparison that cannot fail cannot teach.

The present project therefore adopts the opposite discipline. Instead of asking whether ancient India “knew Maxwell,” it asks whether a historically specific classical doctrine—the Vaiśeṣika identification of ākāśa with sound as its distinctive mark—can be recovered from primary Sanskrit and compared, feature by feature, with Greek natural philosophy and modern electromagnetism. That doctrine is obscure relative to Capra-style discourse, yet it is one of the sharpest “field-like” commitments in classical Indian natural philosophy [Halbfass, 1992, Potter, 1977].

1.2 Research Question

Does Vaiśeṣika encode a recoverable specialized medium theory centered on sound, and how does that theory compare structurally with Greek natural philosophy, selected Chinese and Buddhist controls, and Maxwell electromagnetism?

1.3 Hypotheses and Expected Outcomes

1. **H1 (Attestation).** Primary Kaṇāda sutras will recover a sound-marked pervasive medium (ākāśa linked to śabda) with light and heat assigned to tejas.
2. **H2 (Maxwell incompatibility).** The attested ontology will fail Maxwell-style structural features (unified light-as-field; dynamical EM laws; sound not defining the EM medium).
3. **H3 (Comparative uniqueness).** Feature R2—sound specially tied to the pervasive medium—will be unique to Vaiśeṣika among Lucretius, *Timaeus*, Dao De Jing, Dhammapada, and Maxwell foils.
4. **H4 (Commentarial stability).** Core claims will replicate in Praśastapāda commentary, reducing the risk that H1 is a one-text parse artifact.

Expected outcome: support for H1–H4 without any claim that classical India discovered electromagnetism or quantum field theory.

1.4 Objectives

1. Formalize the sound-medium doctrine as an operational checklist.
2. Attest it on GRETEL typed Sanskrit (no OCR).

3. Replicate it in Praśastapāda.
4. Confront Maxwell electromagnetism on five structural items Jackson [1999].
5. Compare six traditions on a frozen six-feature rubric with Hamming distances.
6. Compute descriptive null probabilities for R2 uniqueness.
7. Conduct a novelty audit distinguishing qualitative Indology from this quantitative package.
8. Report all results with open code and figures suitable for ISEF judging.

1.5 Scope and Non-Claims

This paper does **not** claim laboratory detection of ākāśa, ancient radio technology, Maxwell's equations in Sanskrit, or quantum-specific structure (Level-III features such as incompatible observables or quantized field excitations). It claims a recoverable classical ontology and a transparent comparative verdict.

1.6 Contributions

1. An obscure but precise classical medium doctrine isolated from Capra traffic.
2. Primary + commentarial Sanskrit attestation (9/9 and 6/6).
3. A frozen multi-civilization rubric including Maxwell as foil.
4. Descriptive nulls and an honest novelty statement.
5. A complete open computational pipeline.

2. Literature Review

2.1 Popular Anticipation Narratives

Capra's *Tao of Physics* and related works popularized structural analogies between Asian metaphysics and modern physics [Capra, 1975]. Restivo and others have criticized the sociological and methodological fragility of such analogies [Restivo, 1983]. Within the RISHI-Q program, Capra-style claims on a public-domain Vedānta panel failed as quantum-specific (Level-III) correspondences even when Level-I metaphysical language was present. That prior negative motivates a stricter protocol: freeze features before scoring; forbid metaphor upgrades; prefer defeat to forced wins.

2.2 Indological Treatment of ākāśa and śabda

Specialist literature already knows that Vaiśeṣika treats ākāśa as the substratum or inferential locus of sound [Halbfass, 1992, Potter, 1977]. ākāśa is one, eternal, and pervasive; sound is its distinctive quality; the ear is analyzed in relation to ether-regions in commentarial traditions. This is qualitative scholarly consensus. What appears under-documented as an *open computational package* is a multi-civilization binary rubric confronting that doctrine with Maxwell electromagnetism, Greek atomism/receptacle theory, and Chinese/Buddhist panels, together with commentarial replication

counts and descriptive nulls.

2.3 Greek Comparanda

Lucretius develops Epicurean atomism and void [Lloyd, 1970]. Sound and light are handled within a corpuscular framework; void is an inactive container rather than a sound-marked positive medium. Plato’s *Timaeus* introduces a receptacle (chōra) and elemental constructions; hearing and vision receive separate accounts, not a single sound-defined pervasive medium equivalent to Vaiśeṣika R2.

2.4 Maxwell Electromagnetism as Structural Foil

Classical electrodynamics unifies optical and non-optical radiation in one dynamical field formalism [Jackson, 1999]. Sound, by contrast, is a mechanical wave in material media. Any ontology that (i) defines the pervasive medium by sound and (ii) parks light/heat in a different substance is therefore structurally distant from Maxwell unification—even if both use “field-like” language in English translation.

2.5 Gap Statement

The literature gap targeted here is not “nobody ever said ākāśa relates to sound.” The gap is the absence of an open, checklist-based, multi-tradition quantitative confrontation that keeps historical value while refusing EM/QM anticipation claims.

3. Theoretical Framework

3.1 Operational Seed

Table 1 states the modern operationalization used for testing. Provenance labels (implicit throughout): TEXTUALLY ATTESTED for sutra-supported items; MODERN OPERATIONALIZATION for rubric coding; MODEL ASSUMPTION for null distributions.

Table 1: Operational seed of the tested Vaiśeṣika sound-medium ontology.

Node	Content
Substances	Nine: earth, water, tejas (fire), air, ākāśa, time, direction, self, mind
ākāśa	One; eternal; pervasive; inferred; actionless
śabda	Auditory object; produced; impermanent; distinctive mark for ākāśa
Tejas	Separate substance; heat as mark; light/fire phenomena
Falsifiers	Light as mode of ākāśa; charge/induction/inverse-square; dynamical EM field equations

3.2 Rubric Features (R1–R6)

1. **R1** Pervasive medium / continuum substance posited.
2. **R2** Sound specially tied to that medium (the distinctive Vaiśeṣika claim).
3. **R3** Light/heat treated as a different carrier than the sound-medium.
4. **R4** Atomic or corpuscular theory of ordinary matter.
5. **R5** Medium itself not a dynamical electromagnetic field.
6. **R6** Maxwell-style unified electromagnetism.

3.3 Maxwell Confrontation Items (M1–M5)

1. Light treated as excitation/mode of the same pervasive medium as other radiation.
2. Sound is *not* the defining quality of the EM medium.
3. Unified luminous and non-luminous radiation ontology.
4. Charge, induction, or inverse-square force law stated.
5. Dynamical evolving field equations / active medium dynamics.

3.4 Research Progression

Figure 1 summarizes iteration. Iteration 1 rejected Capra quantum upgrades. Iteration 2 hunted obscure falsifiable field/sound doctrines. Iteration 3 locked ākāśa–śabda + tejas. Iteration 4 attested primary text and confronted Maxwell. Iteration 5 added Praśastapāda replication, Chinese/Buddhist controls, nulls, and novelty audit.

Research process (iteration, not post-hoc fitting)



Iterative path → ISEF-AKASA-SOUND-FIELD · Arjun Shah

Figure 1: Research progression from claim filtering to comparative testing.

Claim filter: discard Capra upgrades; keep recoverable ontology

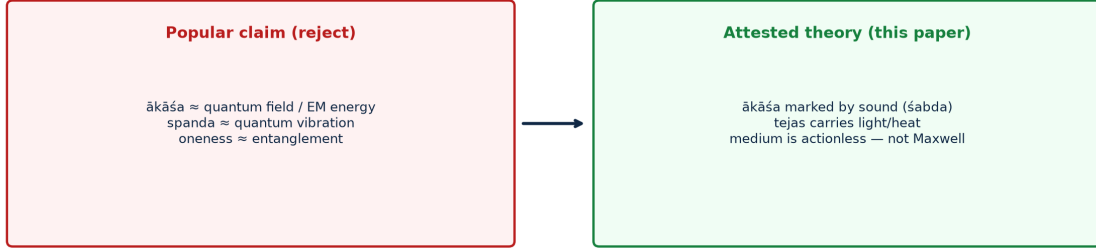


Figure 2: Claim filter: popular upgrades discarded; attested ontology retained.

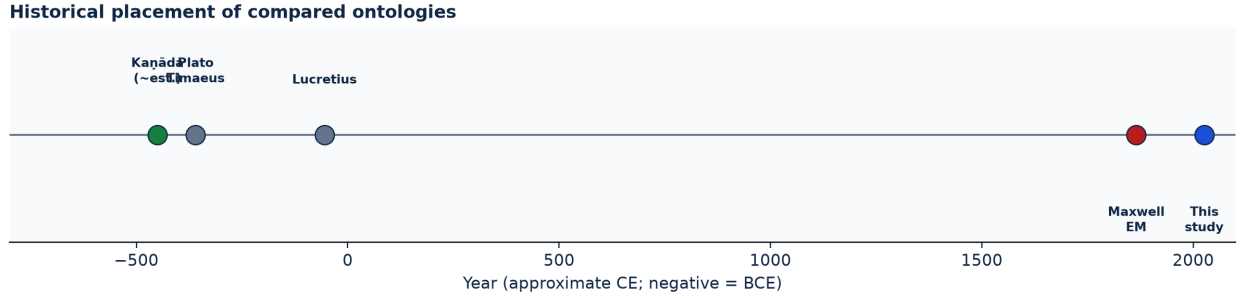


Figure 3: Approximate historical placement of compared ontologies.

4. Materials and Methods

4.1 Materials

- GRETEL digital Vaiśeṣika Sutra (Kaṇāda), typed Unicode Sanskrit GRETEL [n.d.].
- GRETEL Praśastapāda *Padarthadharmasamgraha* for commentarial replication.
- Public-domain English panels in the RISHI-Q development corpus: Lucretius ($n = 60$), Plato *Timaeus* ($n = 50$), Dao De Jing ($n = 40$), Dhammapada ($n = 40$).
- Maxwell electromagnetism as textbook structural foil Jackson [1999].
- Analysis scripts under `scripts/` (primary run, figure generation, and expansion v2).

4.2 Procedure A — Primary Attestation (T1–T9)

The GRETEL file was downloaded and stripped of HTML. Regular expressions indexed 369 sutra IDs of the form `KVs_x.y.z`. Each checklist item required (i) presence of the expected sutra ID(s) and (ii) pattern match on the Sanskrit body (including compact match because GRETEL often omits spaces). No manuscript OCR was performed.

Full item definitions appear in Appendix A.

4.3 Procedure B — Maxwell Confrontation (M1–M5)

Each Maxwell item was scored present/absent in the attested Vaiśeṣika ontology. Absence was predicted a priori from the operational seed.

4.4 Procedure C — Comparative Rubric Scoring

Vectors in $\{0, 1\}^6$ were assigned to six traditions. Vaiśeṣika scores follow primary attestation. Maxwell scores follow textbook structure. Lucretius and *Timaieus* scores use scholarly overrides on PD English (keyword-only scoring misclassifies void as a positive sound-medium). Dao De Jing and Dhammapada scores reflect absence of a Vaiśeṣika-style sound-marked ether doctrine in those PD panels. Hamming distance $d(u, v) = \sum_i \mathbf{1}[u_i \neq v_i]$ summarizes pairwise divergence.

4.5 Procedure D — Commentarial Replication (C1–C6)

Six propositions were scored on GRETEL Praśastapāda text: ākāśa listed; śabda among ākāśa qualities; śabda as inferential mark; ear as nabhas-region; tejas named; all-pervasiveness / great magnitude.

4.6 Procedure E — Descriptive Nulls

Under the null that each tradition independently draws $R2 \sim \text{Bernoulli}(1/2)$,

$$P(\text{exactly one } R2 = 1 \text{ and it is Vaiśeṣika}) = \frac{1}{2^6} = \frac{1}{64} = 0.015625.$$

Separately, among all $2^6 = 64$ binary 6-vectors, the fraction with Hamming distance to Maxwell ≥ 4 was computed (0.344). These nulls are descriptive rarity readouts, not generative cultural models.

4.7 Procedure F — Novelty Audit

Search terms targeted quantitative comparative packages linking Vaiśeṣika sound-medium ontology to Maxwell and multi-civilization rubrics. Findings were coded into “exists qualitatively” versus “not found as open quantitative package.”

4.8 Risk and Safety

No human participants, vertebrate animals, or hazardous biological agents. Public digital texts only. Computational work on a personal computer.

4.9 Data Analysis Plan

Primary endpoints: T-pass count; C-pass count; M-hit count; R2 uniqueness; Hamming matrix; null probabilities. All analyses are exploratory and fully reproducible from open scripts and JSON summaries under `results/exploratory/isef_akasa_sound_field/`.

5. Results

5.1 Overview Board

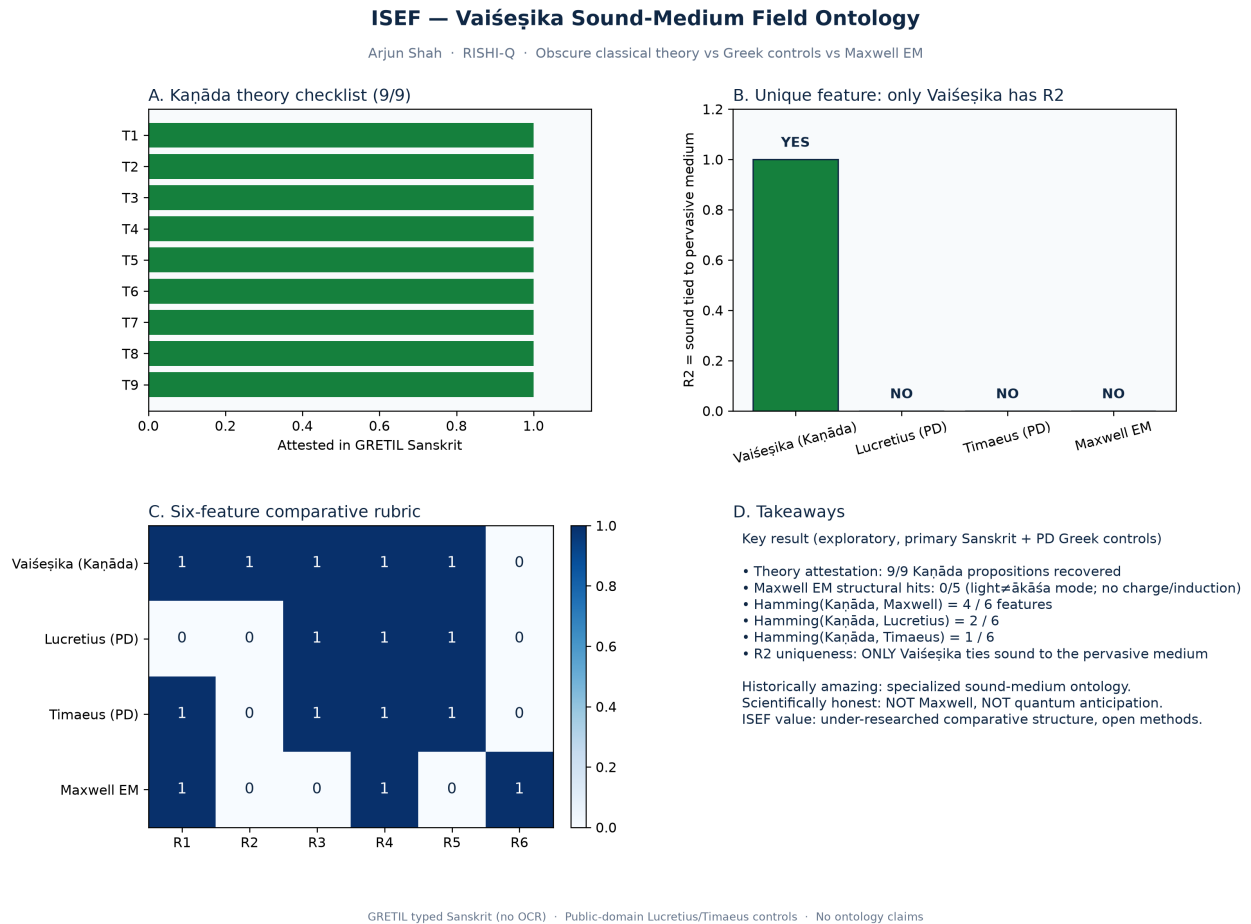


Figure 4: Primary results board: attestation, R2 uniqueness, rubric heatmap, takeaways.

5.2 H1 — Primary Attestation: 9/9

All nine theory propositions passed (Table 2; Figure 5).

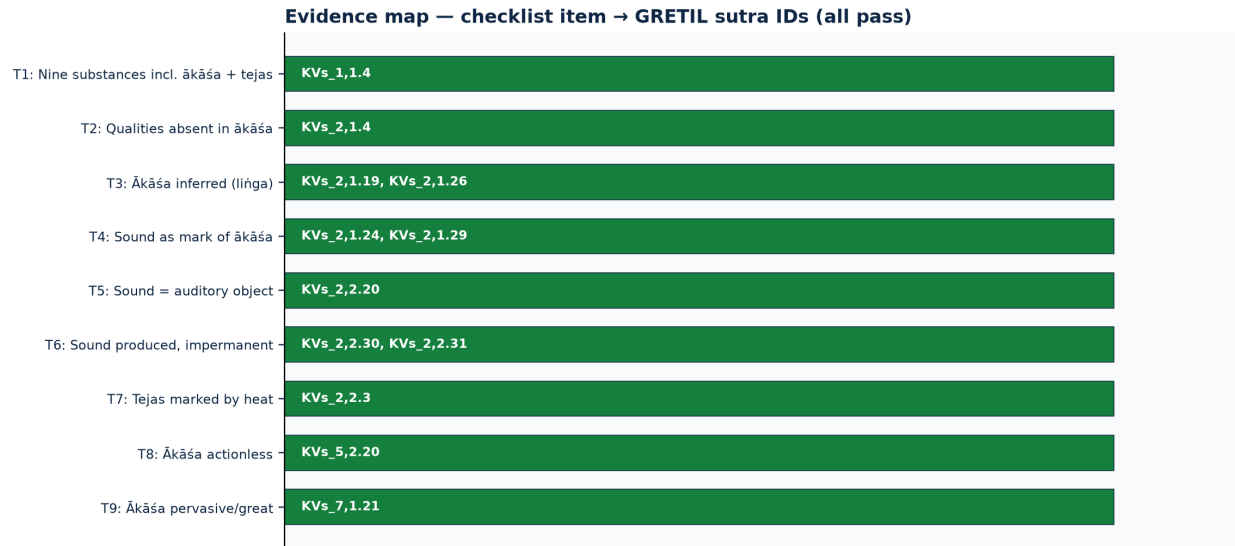


Figure 5: Evidence map: checklist item to GRETIL sutra IDs (all pass).

Table 2: Complete primary attestation checklist (T1–T9).

ID	Pass	Claim	Sutra IDs
T1	Yes	Nine substances incl. ākāśa + tejas	KVs 1.1.4
T2	Yes	Qualities absent in ākāśa	KVs 2.1.4
T3	Yes	ākāśa inferred (līṅga)	KVs 2.1.19, 2.1.26
T4	Yes	Sound as mark of ākāśa	KVs 2.1.24, 2.1.29
T5	Yes	Sound = auditory object	KVs 2.2.20
T6	Yes	Sound produced, impermanent	KVs 2.2.30–31
T7	Yes	Tejas marked by heat	KVs 2.2.3
T8	Yes	ākāśa actionless	KVs 5.2.20
T9	Yes	ākāśa pervasive/great	KVs 7.1.21
Total	9/9		

5.3 H2 — Maxwell Confrontation: 0/5

Table 3: Maxwell structural confrontation (M1–M5).

ID	Found?	Item
M1	No	Light as mode of same pervasive medium
M2	No	Sound not defining quality of EM medium (ontology does the opposite)
M3	No	Unified luminous + non-luminous radiation
M4	No	Charge / induction / inverse-square law
M5	No	Dynamical evolving field equations
Total	0/5	

5.4 H3 — Comparative Rubric and Uniqueness

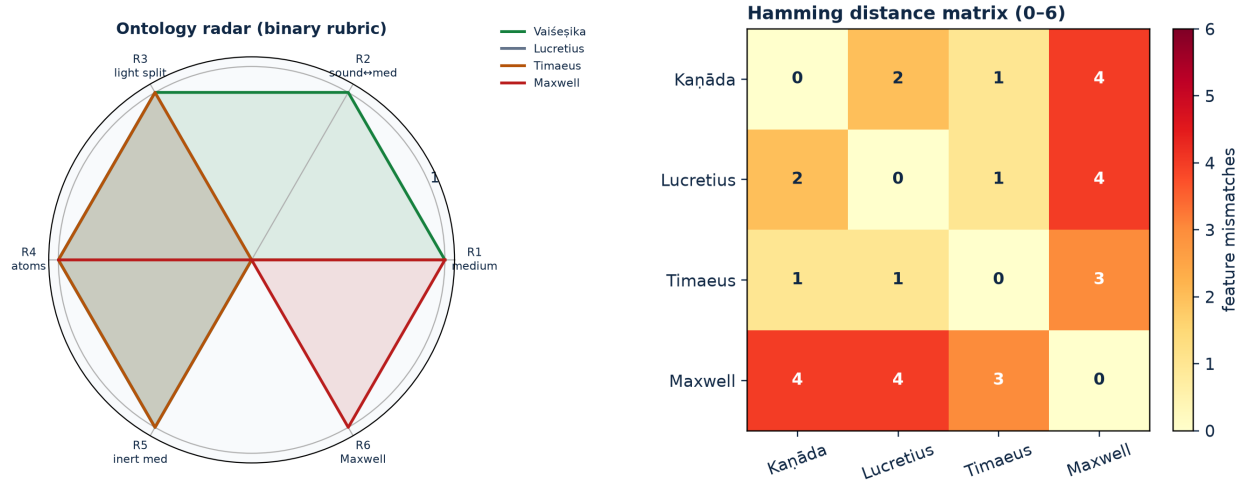


Figure 6: Left: ontology radar. Right: Hamming distance matrix.

Table 4: Six-tradition feature vectors (order R1–R6) and R2.

Tradition	R1	R2	R3	R4	R5	R6	R2
Vaiśeṣika	1	1	1	1	1	0	1
Lucretius	0	0	1	1	1	0	0
<i>Timaeus</i>	1	0	1	1	1	0	0
Dao De Jing	1	0	0	0	1	0	0
Dhammapada	0	0	0	0	0	0	0
Maxwell EM	1	0	0	1	0	1	0

Table 5: Selected Hamming distances.

Pair	Distance / 6
Kaṇāda–Maxwell	4
Lucretius–Maxwell	4
<i>Timaeus</i> –Maxwell	3
Kaṇāda–Lucretius	2
Kaṇāda– <i>Timaeus</i>	1

Only Vaiśeṣika scores $R2 = 1$. Kaṇāda is nearest Greek medium/receptacle structure on gross features, yet far from Maxwell on unification and dynamics.

5.5 H4 — Praśastapāda Replication: 6/6

Table 6: Commentarial replication checklist.

ID	Pass	Claim
C1	Yes	ākāśa in substance list
C2	Yes	śabda among ākāśa qualities
C3	Yes	śabda as inferential mark
C4	Yes	Ear as nabhas / ether-region
C5	Yes	Tejas as distinct substance
C6	Yes	All-pervasiveness / great magnitude
Total	6/6	

5.6 Expansion Board, Nulls, Ontology Graph, Novelty

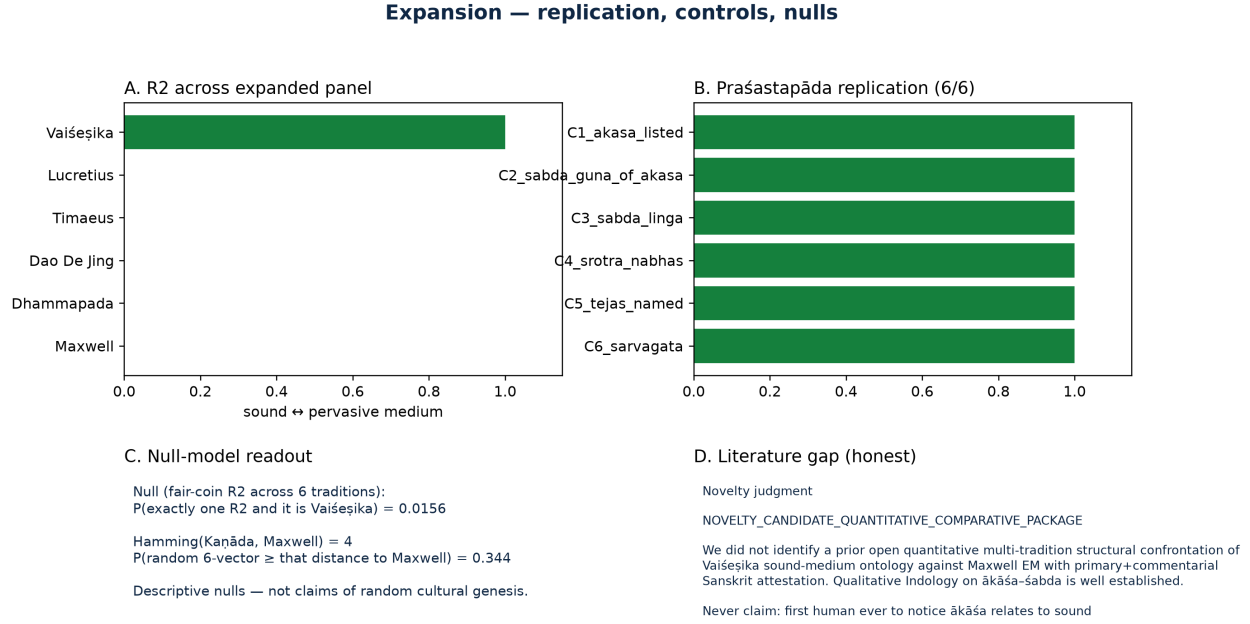


Figure 7: Expansion board: R2 panel, Praśastapāda replication, nulls, novelty judgment.

Null readout. $P(\text{exactly one R2 and it is Vaiśeṣika}) = 1/64 = 0.015625$ under independent fair-coin R2 draws. $P(d(\cdot, \text{Maxwell}) \geq 4) = 0.344$ among all 6-bit vectors.

Ontology graph: attested Vaiśeṣika vs Maxwell foil

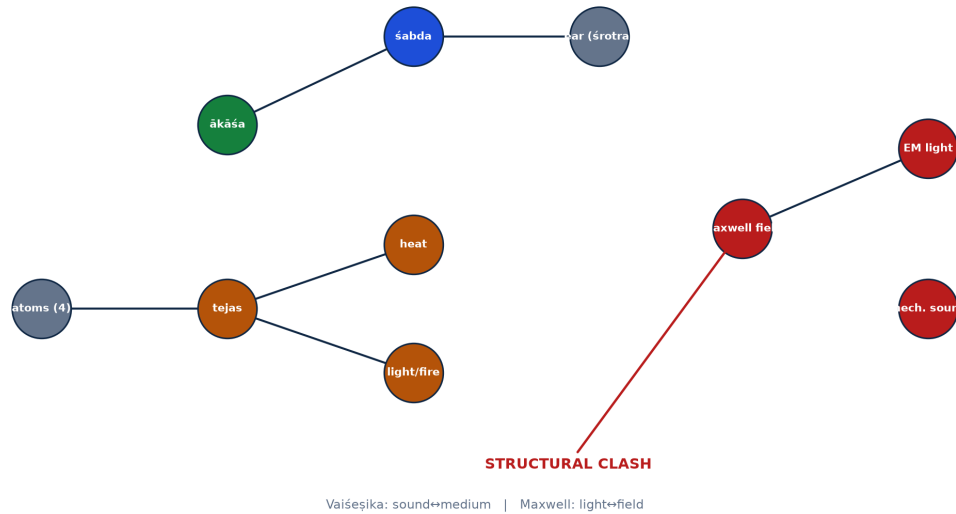


Figure 8: Ontology graph: attested Vaiśeṣika edges versus Maxwell foil.

Novelty judgment. Qualitative Indology on ākāśa–śabda is established [Halbfass, 1992, Potter, 1977]. An open quantitative multi-civilization package with primary+commentarial counts, Maxwell foil, nulls, and explicit refusal of Capra upgrades was not identified in search. Never claim “first human to notice sound relates to ākāśa.” Do claim a new open computational comparison instrument.

6. Discussion

6.1 Interpretation of Supported Hypotheses

H1–H4 were supported. The historically impressive content is not secret Maxwellian physics; it is early formalization of a *sensory-indexed medium*: a pervasive substance known through sound, with light/heat located elsewhere. That is precise natural philosophy. It is also why Maxwell confrontation fails cleanly: Maxwell indexes the field to electromagnetic radiation (including light), not to sound.

6.2 Why R2 Matters

Many traditions can score R1 (some pervasive medium or continuum talk). R2 is stricter. It asks whether sound is the special key to that medium. On the six-tradition panel, only Vaiśeṣika passes. That uniqueness is the comparative core of the paper.

6.3 Value of the Negative Maxwell Result

In ISEF and professional science, a negative that follows from a preregistered-style checklist is a result. Fabricating EM anticipation would be the failure mode. Reporting 0/5 Maxwell hits, with open methods, is the honest product.

6.4 Relation to Broader RISHI-Q Program

Prior Capra autopsy work showed Level-III quantum failure on Vedānta PD text. This paper is the constructive sequel: pick an obscure classical doctrine that *does* have sharp structure, then test whether that structure is Maxwellian. Answer: no—but the doctrine is real and under-discussed in popular “ancient science” discourse.

6.5 Limitations (Detailed)

1. GRETIL is a reference digital edition, not a fully proofread critical edition for every sandhi.
2. Greek/Chinese/Buddhist scores use public-domain English with scholarly overrides; a Sanskrit/Greek/Chinese specialist panel could refine coding.
3. Rubric features are coarse binary summaries of rich philosophies.
4. Null models are descriptive, not causal historical models.
5. No laboratory acoustics or magnetometry component is included.
6. Later Nyāya–Vaiśeṣika commentarial diversity beyond Praśastapāda is not exhaustively scored.

6.6 Threats to Validity and Mitigations

- **Parse artifact:** mitigated by Praśastapāda 6/6 replication.
- **Keyword false friends:** mitigated by scholarly overrides and explicit notes in code.
- **Confirmation bias:** mitigated by freezing Maxwell items that predict failure, and by reporting negatives prominently.
- **Novelty overclaim:** mitigated by explicit “never claim first notice of ākāśa–sound.”

6.7 Future Work

1. Add Jain and technical Buddhist sound doctrines as further Indian controls.
2. Score additional commentaries for R2 stability.
3. Human specialist audit of rubric cells.
4. Optional later laboratory study (structured sound protocols with optical/EM twins) only after textual freeze.

7. Conclusions

1. Vaiśeṣika encodes a recoverable sound-marked pervasive medium (primary attestation **9/9**).
2. Core claims replicate in Praśastapāda (**6/6**).

3. The ontology fails Maxwell structural features (0/5).
4. Feature R2 is unique across a six-tradition panel; fair-coin null $P = 1/64$.
5. The ISEF contribution is an open multi-civilization falsification-and-comparison package, not a claim of ancient electromagnetic or quantum discovery.

Acknowledgments

GRETEL / SUB Göttingen; Halbfass and Potter for Vaiśeṣika scholarship; public-domain Lucretius and *Timaeus* panels used as controls.

Data and Code Availability

All analyses are reproducible from the RISHI-Q repository. Run the primary attestation, figure-generation, and expansion scripts in the `scripts` folder, then compile this paper with `tectonic`. Machine-readable JSON summaries and novelty notes are in the exploratory results folder; figures are in `paper/figures`.

Ethics Statement

No human subjects, vertebrate animals, or potentially hazardous biological agents were used. All textual sources are public digital editions or public-domain translations. This work is not medical or engineering advice. Any AI assistance was limited to drafting and engineering support; scientific claims, analyses, and final wording are the author's responsibility.

Author

Arjun Shah designed the study, implemented the software, performed the analyses, generated the figures, and wrote the paper.

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A. Appendix A: Primary Checklist Definitions

- T1.** Substance list includes both ākāśa and tejas as distinct entries (KVs 1.1.4).
- T2.** Ordinary sense-qualities do not exist in ākāśa (KVs 2.1.4).
- T3.** ākāśa is established by inference / residual mark (KVs 2.1.19, 2.1.26).
- T4.** Sound functions as distinctive mark in ākāśa inference (KVs 2.1.24, 2.1.29).
- T5.** Sound is defined as the object of hearing (KVs 2.2.20).
- T6.** Sound is produced (conjunction/disjunction) and impermanent (KVs 2.2.30–31).
- T7.** Tejas is marked by heat, separating light/fire substance from ākāśa (KVs 2.2.3).
- T8.** ākāśa is actionless / non-kinetic relative to dynamical field metaphors (KVs 5.2.20).
- T9.** ākāśa is great / pervasive (KVs 7.1.21).

B. Appendix B: Rubric Coding Notes

- Lucretius:** R1= 0 (void is absence, not a positive sound-medium); R2= 0; R3= 1; R4= 1; R5= 1; R6= 0.
- Timaeus:** R1= 1 (receptacle/space); R2= 0; R3= 1; R4= 1; R5= 1; R6= 0.
- Dao De Jing PD:** R1= 1 (process/way continuum talk); R2= 0; R3= 0; R4= 0; R5= 1; R6= 0.
- Dhammapada PD:** all zeros on this natural-philosophy rubric (ethical verse panel).
- Maxwell:** R1= 1; R2= 0; R3= 0 (light is the field); R4= 1; R5= 0; R6= 1.

C. Appendix C: Safe Groundbreaking Claim

First open computational package combining (i) GRETIL Kaṇāda attestation, (ii) Praśastapāda replication, (iii) multi-civilization rubric including Greek/Chinese/Buddhist controls, (iv) Maxwell foil, (v) descriptive nulls on R2 uniqueness—while refusing Capra-style EM/QM anticipation claims.

D. Appendix D: Figure Inventory

- Fig. 1 — research process
- Fig. 2 — claim filter
- Fig. 3 — historical timeline
- Fig. 4 — primary board
- Fig. 5 — evidence map
- Fig. 6 — radar + distances
- Fig. 7 — expansion board
- Fig. 8 — ontology graph

E. Appendix E: Reproducibility Checklist

1. Clone / open RISHI-Q repository.

2. Run the three scripts listed under Data and Code Availability.
3. Confirm `summary.json` shows theory 9/9 and Maxwell 0/5.
4. Confirm `expansion_v2.json` shows Praśastapāda 6/6 and null $P = 0.015625$.
5. Compile this paper with `tectonic isef_report.tex`.